Build your own amplifier

Nash David
nash.david@thinkdigit.com

DIYs are fun. Electronics is fun too, but the thought of putting your circuitry together and soldering them seems like a daunting task. If done right, you should be able to get your circuitry up and running in the first go. But as an amateur, mistakes are bound to happen. You might need to take a closer look to ensure all the connections are right. We'll attempt building an amplifier around the TDA 2030A audio power amplifier IC. We referred to the data sheet of this IC and opted for the bridge amplifier configuration for its simplicity and price.

Although the circuit looks small, building it can be a tough nut to crack for an amateur. Taking care of a few things, however, could take you a long way. This way you'll find it really easy.

What you need

Resistors
R1, R2, R3, R5, R7 – 22k
R4, R6 – 680E
R8, R9 – 1E

Capacitors
C1 – 2.2uF
C2, C6 – 100uF
C3, C7 – 100nF
C4, C5 – 22uF
C8, C9 – 0.22uF

ICs
7815, 7915 – Positive and negative voltage regulators (15V)
TDA2030A x 2.

18-0-18V step-down transformer
3.5-mm audio jack (male)
Connecting wire
Multimeter
Soldering iron and alloy
General purpose PCB
Speaker connector

Your components

Resistors are thin components with coloured bands. These coloured bands are standard convention used in reading the values of resistors.

Among the various types of capacitors, you'll use ceramic as well as electrolytic capacitors while building this circuit. The difference between the two is that electrolytic capacitors have polarised terminals while ceramic capacitors don't.

You can connect ceramic capacitors irrespective of polarity, but when soldering electrolytic capacitors, ensure the terminals are oriented across appropriate voltage potentials. Also, electrolytic capacitors have a voltage rating. As a rule, opt for a voltage rating of double the power supply.

Effectively, for this circuit, with a power supply of 15 V, use electrolytic capacitors with a voltage rating exceeding 30 V. We opted for 62 V.

Integrated circuits have a pin-out diagram and you should refer to the appropriate data sheet for information on pin-outs to avoid damage to the component as a result of wrong connections.

Circuit diagram

Understanding circuit schematics and following connections is vital in getting your amplifier to work. The circuit diagram in Figure 1, A line indicates a connection path. Similarly, two intersecting lines mean the connections don't intersect each other. If the two lines intersect each other and have a dot at the point of intersection then the two paths are connected. Accordingly, positive supply voltage is connected across pin 5 of the two ICs and pins 1 and 5 of IC1 are not connected to each other. These are some common errors while building a circuit.

As you can see in the circuit diagram, there are two amplifier ICs with an identical circuitry for each stage. This supporting circuitry is wired such that both the ICs handle the signal identically. However, one generates a positive signal while the second stage generates a negative signal, thereby increasing the resultant potential difference of the output signal. This way, you have double the output with the same amplification factor. This way of wiring an amplifier is known as bridging. A sample of the input signal is fed to the inverting input of the amplifier to get an inverted output. The output of each stage is then connected to the speaker terminals.

This circuit needs a symmetrical power supply built around a centre-tapped transformer (18-0-18V/1A). We got one for around Rs. 90. The two wires on one end of the transformer (known as the primary winding) should be connected to the AC 230 V supply. Similarly, in the case of the centre-tapped transformer, you'll find three wires at the other end (secondary winding). The wire at the centre should be considered as ground.

If you take a multimeter, set it to AC voltage measurement (indicated by the V~ symbol) and measure the voltage across the terminals (the centre terminal should be reference), you will get a reading around 18-20V. However, the two AC

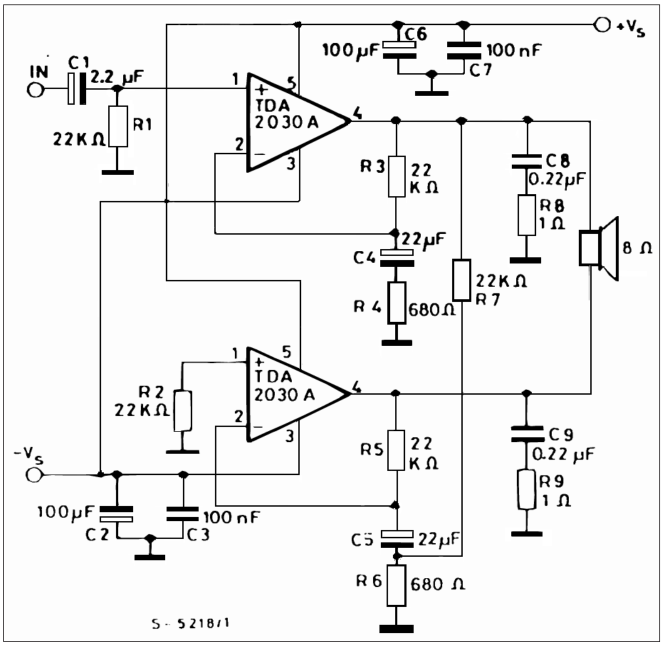

Figure 1: TDA 2030A wired in a bridge configuration can give you a higher audio output with fewer components or amplification stages